

4Layers IoT — Master AWS Infrastructure, Cost & Architecture Report

Current Active Setup + AWS IoT Core Scaling Comparison + \$120 Credit Utilization Roadmap

EXECUTIVE SUMMARY (Master Summary): Is single report mein 4Layers Smart Home System ka **Current Live AWS Setup, \$120.00 AWS Free Credits Status, Current EMQX vs AWS IoT Core Comparison**, aur **100 to 1,000 Device Scaling Roadmap** poora shamil hai. Current active \$120.00 AWS Free Credits se **11-12 Mahine tak CLIENT POCKET SPEND RS. 0.00 (Zero Rupees)** rahega.

1. Current Live Active Infrastructure (Abhi AWS Par Kya Active Hai)

System Component	Active Service / Technology	Live Endpoint / Configuration Details	Monthly Status
Backend API Service	AWS App Runner (FastAPI Docker)	https://edabtyrnpvy.ap-south-1.awsapprunner.com	Active (\$120 Credits)
Database Engine	AWS RDS PostgreSQL (db.t4g.micro)	PostgreSQL 15 Single-AZ with Encrypted Storage	Active (\$0 Free Tier)
MQTT IoT Broker	EMQX Cloud Serverless (TLS Port 8883)	i26a1c71.ala.asia-southeast1.emqxsl.com	Active (\$0 Spend Cap)
Voice Assistant OAuth	FastAPI Built-in OAuth 2.0 Server	/oauth/authorize & /oauth/token	Active & Live
Google Home Fulfillment	Action.Devices Smart Home API	/api/google/fulfillment (SYNC, QUERY, EXECUTE)	Active & Live
ESP32 Firmware	Production Firmware v2.0	NVS Memory State Save + 60s KeepAlive + 1024 TLS Buffer	Active (Zero Toggles)
Mobile App	React Native (Expo) Release APK	4layers-v1.0-release.apk (Size: 34.1 MB)	Built & Ready

2. Current EMQX Cloud vs AWS IoT Core Comparison

Feature / Parameter	Current EMQX Cloud Setup	AWS IoT Core Setup	Fayda / Winner
ESP32 Firmware Coding	Simple PubSubClient + Username/Password. Lightweight.	Har board mein alag X.509 SSL Certificate burn karna padega.	EMQX (Easy Code)
Factory Mass Production	Same single firmware binary sabhi 100 boards par flash ho jata hai.	Har single board ke liye AWS Console se alag cert download karna hoga.	EMQX (Fast Production)
Bill Safety & Risk	100% Hard Spend Cap (\$0.00) active. Extra bill aane ka 0% chance.	Metered Billing per msg & per min connectivity.	EMQX (Zero Risk)
Live Dashboard Monitor	Ready-made web console mein connections & topics live dikhte hain.	CloudWatch plain text logs milte hain.	EMQX (Live Dashboard)
Scaling Capacity	1 Million Free Msgs/mo limit (Spend Cap active).	Unlimited scaling (Laakhon devices ek saath).	AWS IoT (Big Scale)

3. Current AWS Free Credits Status (\$120.00 Credits Pool)

Credit Metric	Amount (USD)	Amount (INR)	Status & Validity
Total Issued AWS Credits	\$120.00	■10,020 INR	Active (Expires 28 July 2027)
Abhi Tak Ka Total Usage	\$1.38	■115 INR	Deducted from Free Credits
REMAINING FREE CREDITS	\$118.62	~■9,905 INR	Available for Next 11 Months
CLIENT POCKET SPEND (MONTHS 1-11)	\$0.00	■0.00 INR	100% FREE FOR 11 MONTHS

4. 100 Devices vs 1,000 Devices Scaling & Profitability Model

Scale Metric	100 Devices (Abhi Ka Scale)	1,000 Devices (Future Scale)	Unit Cost & Profit
Server Cost (EC2 + Storage)	\$5.00/mo (■415)	\$8.80/mo (■735)	Shared Server Capacity
MQTT Messaging & Connectivity	\$6.11/mo (■510)	\$61.05/mo (■5,088)	Scaled Bandwidth
TOTAL MONTHLY SYSTEM SPEND	~\$11.11/mo (~■925/mo)	~\$69.85/mo (~■5,820/mo)	■5.82 per home / mo
1,000 HOMES PROFITABILITY	Covered 100% by \$120 credits	Subscription @ ■50/mo = ■50,000	+■44,180/mo PROFIT!

5. Year 2 Roadmap (Credits Khatam Hone Ke Baad)

Year 2 Plan: \$120 credits khatam hone ke baad (11-12 mahine baad), hum system ko Reserved Single EC2 Docker Server (\$3.50/mo = ~\$290/mo) par shift kar denge. Total system running cost hamesha ke liye **\$350 se \$400/month** par lock rahega!